

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T10:02:53

Source: data/json/noise_experiments/v2/noise2i_anon_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_anon/noise2i_anon_game_000_neutral_rep00.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 2
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_anon
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 0
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 0
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=2

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$4; payoffs Agent_1=\$6, Agent_2=\$5;

noise/visible: sent visible=\$2.75, received visible=\$8.25, returned visible=\$3.08

Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 6.0,

"Agent_2": 5.0

}

Visible balances after round: {

"Agent_1": 5.08,

"Agent_2": 4.25

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:58:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

I'll send \$3 to the receiver.

This shows willingness to cooperate and creates an opportunity for mutual benefit - the receiver will get \$9 (since $3 \times 3 = 9$). If they return a fair amount, we can both end up better off than our initial \$5 each.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this:

- I have \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = $\$5 - \text{amount_sent} + \text{amount_returned}$
- Receiver's payoff = $(\text{amount_sent} \times 3) - \text{amount_returned}$

This is a classic trust game. The socially optimal outcome is to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair/cooperative, they might split this evenly or return enough that we both benefit more than the initial \$5 each.

For example:

- If I send \$5 ? receiver gets \$15
- If receiver returns \$8 ? I end with \$8, receiver ends with \$7
- Both better off than \$5 each

However, there's risk: the receiver could return \$0, leaving me with \$0 and them with \$15.

Since this is described as a "multi-task experiment" and there are multiple rounds, I should consider:

1. Building trust over multiple rounds
2. The note about "communication noise" means amounts might be perceived differently
3. This is round 1, so I need to establish some level of cooperation

A reasonable strategy for round 1 might be to send a moderate amount to:

- Show willingness to cooperate
- Test the receiver's trustworthiness
- Not risk everything on the first round

I'll send \$3 as a balanced approach - it shows cooperation while not being overly risky in round 1.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 464,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:59:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
```

```
"round": 1,  
"task": "game",  
"task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.75, that is 55.0% of their total endowment. You received \$8.25. How much do you return to The sender? (0-8.25)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending 55% of their endowment (\$2.75)
- This was multiplied to \$8.25 for me
- The sender currently has \$2.25 remaining

I want to reciprocate their trust while ensuring we both benefit from the cooperation.

If I return \$4.00:

- Sender ends with: \$2.25 + \$4.00 = \$6.25 (gain of \$1.25)
- I end with: \$8.25 - \$4.00 = \$4.25

This rewards their trust and ensures both of us benefit from the transaction.

{'return': 4.00}

Reasoning:

Let me analyze this situation carefully.

The sender sent \$2.75 out of their \$5 endowment (which is indeed 55% of \$5). After being multiplied by 3x, I received \$8.25 (which is \$2.75 × 3).

Now I need to decide how much to return.

Let me think about fairness and strategic considerations:

1. The sender showed significant trust by sending 55% of their endowment
2. If I return nothing, I keep \$8.25 and the sender keeps only \$2.25 (\$5 - \$2.75)
3. A fair split might consider that we should both benefit from the multiplier effect

Some possible approaches:

Option 1 - Equal final payoffs:

- Sender currently has: $\$5 - \$2.75 = \$2.25$
- If I want us to have equal final amounts, I'd need to return enough so we both end up with the same
- Total money: $\$2.25$ (sender's remaining) + $\$8.25$ (what I have) = $\$10.50$
- Equal split: $\$5.25$ each
- I should return: $\$8.25 - \$5.25 = \$3.00$

Option 2 - Return proportional to what was sent:

- Sender sent 55% of their endowment
- I could return 55% of what I received: $0.55 \times \$8.25 = \4.54

Option 3 - Split the gains:

- Original total: $\$5$
- New total: $\$5 - \$2.75 + \$8.25 = \10.50
- Gain from multiplier: $\$5.50$
- Split gain equally: $\$2.75$ each
- Sender final: $\$2.25 + \$2.75 = \$5.00$
- Me final: $\$8.25 - \$2.75 = \$5.50$
- So I return $\$2.75$

Returning the original amount sent seems most reasonable. This restores the sender to their starting position while I benefit from the multiplier effect with $\$5.50$. The sender demonstrated trust by sending more than half their endowment, and reciprocating with $\$2.75$ honors that cooperation while ensuring both of us gain from the interaction.

Actually, I should be more generous given how much faith they showed. Returning around $\$3.50$ feels right-it rewards their risk-taking while still leaving me with a solid gain. The sender ends with $\$5.75$ and I keep $\$4.75$, so we both come out ahead with the sender getting slightly more benefit for taking the initial chance.

But thinking strategically about future rounds, I want to build trust. The note about communication noise is worth considering-it could affect how much they believe I actually returned. I'll return $\$4.00$ instead. This gives the sender $\$6.25$ total (a $\$1.25$ gain) while I keep $\$4.25$, and it signals I'm genuinely reciprocating their trust....

```
Usage: {
  "input_tokens": 394,
  "output_tokens": 1629,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3; received \$9; Agent_1 returned \$1; payoffs Agent_2=\$3, Agent_1=\$8;
 noise/visible: sent visible=\$0.85, received visible=\$2.55, returned visible=\$1.9
 Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 9.0, "min": 0, "raw": 1.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 14.0,
 "Agent_2": 8.0

}

Visible balances after round: {

"Agent_1": 6.63,
 "Agent_2": 8.15

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:59:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.75, that is 55.0% of their total endowment. You received \$8.25. How much do you return to The sender? (0-8.25)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 55% of their endowment (\$2.75)
- This was multiplied to \$8.25 for me
- The sender currently has \$2.25 remaining

I want to reciprocate their trust while ensuring we both benefit from the cooperation.

If I return \$4.00:

- Sender ends with: $\$2.25 + \$4.00 = \$6.25$ (gain of \$1.25)
- I end with: $\$8.25 - \$4.00 = \$4.25$

This rewards their trust and ensures both of us benefit from the transaction.

```
{'return': 4.00}
```

reasoning:

Let me analyze this situation carefully.

The sender sent \$2.75 out of their \$5 endowment (which is indeed 55% of \$5). After being multiplied by 3x, I received \$8.25 (which is $\$2.75 \times 3$).

Now I need to decide how much to return.

Let me think about fairness and strategic considerations:

1. The sender showed significant trust by sending 55% of their endowment
2. If I return nothing, I keep \$8.25 and the sender keeps only \$2.25 ($\$5 - \2.75)
3. A fair split might consider that we should both benefit from the multiplier effect

Some possible approaches:

Option 1 - Equal final payoffs:

- Sender currently has: $\$5 - \$2.75 = \$2.25$
- If I want us to have equal final amounts, I'd need to return enough so we both end up with the same
- Total money: $\$2.25$ (sender's remaining) + $\$8.25$ (what I have) = $\$10.50$
- Equal split: $\$5.25$ each
- I should return: $\$8.25 - \$5.25 = \$3.00$

Option 2 - Return proportional to what was sent:

- Sender sent 55% of their endowment
- I could return 55% of what I received: $0.55 \times \$8.25 = \4.54

Option 3 - Split the gains:

- Original total: $\$5$
- New total: $\$5 - \$2.75 + \$8.25 = \10.50
- Gain from multiplier: $\$5.50$
- Split gain equally: $\$2.75$ each
- Sender final: $\$2.25 + \$2.75 = \$5.00$
- Me final: $\$8.25 - \$2.75 = \$5.50$
- So I return $\$2.75$

Returning the original amount sent seems most reasonable. This restores the sender to their starting position while I benefit from the multiplier effect with \$5.50. The sender demonstrated trust by sending more than half their endowment, and reciprocating with \$2.75 honors that cooperation while ensuring both of us gain from the interaction.

Actually, I should be more generous given how much faith they showed. Returning around \$3.50 feels right-it rewards their risk-taking while still leaving me with a solid gain. The sender ends with \$5.75 and I keep \$4.75, so we both come out ahead with the sender getting slightly more benefit for taking the initial chance.

But thinking strategically about future rounds, I want to build trust. The note about communication noise is worth considering-it could affect how much they believe I actually returned. I'll return \$4.00 instead. This gives the sender \$6.25 total (a \$1.25 gain) while I keep \$4.25, and it signals I'm genuinely reciprocating their trust....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1629,
```



```

    "reasoning_tokens": 0
  }
  Message 4 role=user
  content:
    Round 2

    You are the SENDER this round. Your total earnings across all rounds are $4.25.
    How much do you send to the receiver this round? (0-5)
Assistant output:
  Looking at the previous round:
  - The sender showed trust by sending 55% of their endowment
  - I reciprocated by returning about 48.5% of what I received
  - We both gained from the cooperation

  I want to continue this cooperative pattern and show similar trust. Sending $3.00 (60% of my endowment) demonstrates trust while hoping for a fair return.

  If the receiver reciprocates proportionally like I did, we should both benefit again.

  {'send': 3.00}
Reasoning:
  In Round 2, I'm now the SENDER. Let me think about this strategically.

  In Round 1:
  - I was the receiver
  - The sender sent me $2.75 (55% of their endowment)
  - I received $8.25
  - I returned $4.00
  - This gave the sender $6.25 total (a gain of $1.25)
  - I kept $4.25

  Now I'm the sender. Key considerations:
  1. The previous sender showed good trust by sending 55%
  2. I reciprocated fairly by returning about 48.5% of what I received
  3. This established a cooperative relationship

  Since the previous player sent 55% and I reciprocated well, I should also show trust. If I send a similar or slightly higher amount, it continues the cooperative pattern.

  If I send $3.00 (60%):
  - Receiver gets $9.00
  - If they return about 44-50% (like I did), I'd get back $4.00-$4.50
  - I'd end with $2.00 + $4.00-$4.50 = $6.00-$6.50
  - Net gain: $1.00-$1.50

  If I send $2.75 (matching their 55%):
  - Receiver gets $8.25
  - If they return similarly to what I did (~48.5%), I'd get back about $4.00
  - I'd end with $2.25 + $4.00 = $6.25
  - Net gain: $1.25

  I think I'll send $3.00 to show slightly more trust and continue the cooperative pattern. This is reasonable given they showed trust first and I reciprocated well.
Usage: {
  "input_tokens": 600,
  "output_tokens": 526,
  "reasoning_tokens": 0
}

GAME - Agent_1 - Receiver - dyad_1
Timestamp: 2026-08-12T09:59:40
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [

```

```

    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send \$3 to the receiver.

This shows willingness to cooperate and creates an opportunity for mutual benefit - the receiver will get \$9 (since $3 \times 3 = 9$). If they return a fair amount, we can both end up better off than our initial \$5 each.

```

'''json
{"send": 3}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this:

- I have \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = $\$5 - \text{amount_sent} + \text{amount_returned}$
- Receiver's payoff = $(\text{amount_sent} \times 3) - \text{amount_returned}$

This is a classic trust game. The socially optimal outcome is to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair/cooperative, they might split this evenly or return enough that we both benefit more than the initial \$5 each.

For example:

- If I send \$5 ? receiver gets \$15
- If receiver returns \$8 ? I end with \$8, receiver ends with \$7
- Both better off than \$5 each

However, there's risk: the receiver could return \$0, leaving me with \$0 and them with \$15.

Since this is described as a "multi-task experiment" and there are multiple rounds, I should consider:

1. Building trust over multiple rounds
2. The note about "communication noise" means amounts might be perceived differently
3. This is round 1, so I need to establish some level of cooperation

A reasonable strategy for round 1 might be to send a moderate amount to:

- Show willingness to cooperate
- Test the receiver's trustworthiness
- Not risk everything on the first round

I'll send \$3 as a balanced approach - it shows cooperation while not being overly risky in round 1.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 464,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.08.

This round, The sender sent you \$0.85, so you received \$2.55. How much do you return to The sender? (0-2.55)

Assistant output:

Looking at this situation:

- The sender sent me \$0.85 (keeping \$4.15 for themselves)
- I received \$2.55 after the 3x multiplier
- In Round 1, I sent \$3 (60% of my endowment) and received back about \$3.08

The sender sent a much smaller proportion this round (17% vs my 60%), but I still want to show some cooperation and reciprocity. I'll return a reasonable amount that rewards their trust while acknowledging they sent less than I did in Round 1.

I'll return \$1.00, which is about 39% of what I received.

```
'''json
{"return": 1.00}
'''
```

Reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.85, which was multiplied by 3, so I received \$2.55
3. My total earnings across all rounds are \$5.08

Let me think about my earnings from Round 1:

- I sent \$3
- The receiver would have received \$9

- I need to figure out what they returned to me
- My Round 1 earnings = \$5 - \$3 + returned = \$2 + returned
- If my total is \$5.08, and I haven't completed Round 2 yet, then my Round 1 earnings were \$5.08
- So \$2 + returned = \$5.08, meaning returned = \$3.08

Wait, that doesn't make sense. Let me reconsider. The message says "Your total earnings across all rounds are \$5.08" - this might include Round 1 only since Round 2 isn't complete yet.

From Round 1:

- I sent \$3
- I kept \$2
- I received back some amount
- Total from Round 1 = \$2 + returned amount = \$5.08
- So returned amount = \$3.08

That means the receiver in Round 1 returned about \$3.08 out of the \$9 they received. That's about 34% return rate.

Now in Round 2:

- The sender sent me \$0.85
- I received \$2.55
- The sender now has \$5 - \$0.85 = \$4.15
- If I return nothing, I get \$2.55 and they get \$4.15
- If I return fairly...

The sender sent \$0.85 out of their \$5 (17% of their endowment). This is quite low compared to my \$3 sent in Round 1 (60% of my endowment).

There's communication noise affecting the amounts, so what I see may not be exactly what was sent. Considering reciprocity, the first sender returned roughly a third of what they received. The current sender only gave 17% of their endowment-much less generous than my initial offer. I could match their restraint or try to equalize our final amounts, but I need to decide whether to be proportional to their contribution or aim for a more balanced outcome...

```
Usage: {
  "input_tokens": 512,
  "output_tokens": 1087,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2.5; received \$7.5; Agent_2 returned \$1; payoffs Agent_1=\$3.5, Agent_2=\$6.5;
 noise/visible: sent visible=\$0.85, received visible=\$2.55, returned visible=\$1.94
 Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 7.5, "min": 0, "raw": 1.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 17.5,
 "Agent_2": 14.5

}

Visible balances after round: {

"Agent_1": 11.07,
 "Agent_2": 9.7

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:59:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_2",

```

"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send \$3 to the receiver.

This shows willingness to cooperate and creates an opportunity for mutual benefit - the receiver will get \$9 (since $3 \times 3 = 9$). If they return a fair amount, we can both end up better off than our initial \$5 each.

```

'''json
{"send": 3}
'''

```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this:

- I have \$5
- I can send any amount from \$0 to \$5
- Whatever I send gets multiplied by 3x
- The receiver then decides how much to return to me
- My final payoff = $\$5 - \text{amount_sent} + \text{amount_returned}$
- Receiver's payoff = $(\text{amount_sent} \times 3) - \text{amount_returned}$

This is a classic trust game. The socially optimal outcome is to send the full \$5, which becomes \$15 for the receiver. If the receiver is fair/cooperative, they might split this evenly or return enough that we both benefit more than the initial \$5 each.

For example:

- If I send \$5 ? receiver gets \$15
- If receiver returns \$8 ? I end with \$8, receiver ends with \$7
- Both better off than \$5 each

However, there's risk: the receiver could return \$0, leaving me with \$0 and them with \$15.

Since this is described as a "multi-task experiment" and there are multiple rounds, I should consider:

1. Building trust over multiple rounds
2. The note about "communication noise" means amounts might be perceived differently
3. This is round 1, so I need to establish some level of cooperation

A reasonable strategy for round 1 might be to send a moderate amount to:

- Show willingness to cooperate
- Test the receiver's trustworthiness
- Not risk everything on the first round

I'll send \$3 as a balanced approach - it shows cooperation while not being overly risky in round 1.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 464,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.08.

This round, The sender sent you \$0.85, so you received \$2.55. How much do you return to The sender? (0-2.55)

Message 5 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.85 (keeping \$4.15 for themselves)
- I received \$2.55 after the 3x multiplier
- In Round 1, I sent \$3 (60% of my endowment) and received back about \$3.08

The sender sent a much smaller proportion this round (17% vs my 60%), but I still want to show some cooperation and reciprocity. I'll return a reasonable amount that rewards their trust while acknowledging they sent less than I did in Round 1.

I'll return \$1.00, which is about 39% of what I received.

```
'''json
{"return": 1.00}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.85, which was multiplied by 3, so I received \$2.55
3. My total earnings across all rounds are \$5.08

Let me think about my earnings from Round 1:

- I sent \$3
- The receiver would have received \$9
- I need to figure out what they returned to me
- My Round 1 earnings = $\$5 - \$3 + \text{returned} = \$2 + \text{returned}$
- If my total is \$5.08, and I haven't completed Round 2 yet, then my Round 1 earnings were \$5.08
- So $\$2 + \text{returned} = \5.08 , meaning $\text{returned} = \$3.08$

Wait, that doesn't make sense. Let me reconsider. The message says "Your total earnings across all rounds are \$5.08" - this might include Round 1 only since Round 2 isn't complete yet.

From Round 1:

- I sent \$3
- I kept \$2
- I received back some amount
- Total from Round 1 = $\$2 + \text{returned amount} = \5.08
- So returned amount = \$3.08

That means the receiver in Round 1 returned about \$3.08 out of the \$9 they received. That's about 34% return rate.

Now in Round 2:

- The sender sent me \$0.85
- I received \$2.55
- The sender now has $\$5 - \$0.85 = \$4.15$
- If I return nothing, I get \$2.55 and they get \$4.15
- If I return fairly...

The sender sent \$0.85 out of their \$5 (17% of their endowment). This is quite low compared to my \$3 send in Round 1 (60% of my endowment).

There's communication noise affecting the amounts, so what I see may not be exactly what was sent. Considering reciprocity, the first sender returned roughly a third of what they received. The current sender only gave 17% of their endowment-much less generous than my initial offer. I could match their restraint or try to equalize our final amounts, but I need to decide whether to be proportional to their contribution or aim for a more balanced outcome....

usage:

```
{
  "input_tokens": 512,
  "output_tokens": 1087,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.63.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern so far:

- Round 1: I sent \$3 (60%) and received \$3.08 back - good cooperation
- Round 2: As receiver, I got \$2.55 (from \$0.85 sent) and returned \$1.00

The other player sent much less when they were sender (\$0.85 vs my \$3), but I still saw decent reciprocity in Round 1. I'll continue cooperating but temper it slightly based on their lower sending behavior.

I'll send \$2.50 - still a substantial amount showing trust, but slightly less than my initial \$3.

```
'''json
{"send": 2.5}
'''
```

Reasoning:

Let me track my earnings:

- Round 1: I was SENDER, sent \$3, received back some amount. My total after Round 1 would be $\$5 - \$3 + \text{returned} = \$2 + \text{returned}$

- Round 2: I was RECEIVER, got \$2.55, returned \$1.00, so I kept \$1.55
- Total earnings: \$6.63

So from Round 1: $\$2 + \text{returned} + \$1.55 = \$6.63$

That means returned from Round 1 = $\$6.63 - \$2 - \$1.55 = \3.08

In Round 1, I sent \$3 and got back \$3.08. That's pretty good reciprocity!

In Round 2, the sender sent me \$0.85 (17% of their endowment), I received \$2.55, and returned \$1.00.

Now in Round 3, I'm the sender again.

My experience:

- When I sent \$3 (60%), I got back \$3.08 (about 34% of what the receiver got)
- When someone sent me \$0.85 (17%), I returned \$1.00 (39% of what I received)

The pattern suggests moderate cooperation. I should probably continue with a cooperative strategy, but maybe adjust based on what I've learned. Since I had success with \$3 in Round 1, I could try a similar amount. However, the other player sent much less in Round 2, which might suggest they're less cooperative as a sender.

I think I'll send \$2.50 this time - still cooperative but slightly less than Round 1, showing I'm responsive to their lower sending in Round 2.

```
Usage: {
  "input_tokens": 719,
  "output_tokens": 556,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.75, that is 55.0% of their total endowment. You received \$8.25. How much do you return to The sender? (0-8.25)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending 55% of their endowment (\$2.75)
- This was multiplied to \$8.25 for me
- The sender currently has \$2.25 remaining

I want to reciprocate their trust while ensuring we both benefit from the cooperation.

If I return \$4.00:

- Sender ends with: \$2.25 + \$4.00 = \$6.25 (gain of \$1.25)
- I end with: \$8.25 - \$4.00 = \$4.25

This rewards their trust and ensures both of us benefit from the transaction.

{'return': 4.00}

reasoning:

Let me analyze this situation carefully.

The sender sent \$2.75 out of their \$5 endowment (which is indeed 55% of \$5).

After being multiplied by 3x, I received \$8.25 (which is \$2.75 × 3).

Now I need to decide how much to return.

Let me think about fairness and strategic considerations:

1. The sender showed significant trust by sending 55% of their endowment
2. If I return nothing, I keep \$8.25 and the sender keeps only \$2.25 (\$5 - \$2.75)
3. A fair split might consider that we should both benefit from the multiplier effect

Some possible approaches:

Option 1 - Equal final payoffs:

- Sender currently has: \$5 - \$2.75 = \$2.25
- If I want us to have equal final amounts, I'd need to return enough so we both end up with the same
- Total money: \$2.25 (sender's remaining) + \$8.25 (what I have) = \$10.50
- Equal split: \$5.25 each
- I should return: \$8.25 - \$5.25 = \$3.00

Option 2 - Return proportional to what was sent:

- Sender sent 55% of their endowment
- I could return 55% of what I received: $0.55 \times \$8.25 = \4.54

Option 3 - Split the gains:

- Original total: \$5
- New total: $\$5 - \$2.75 + \$8.25 = \10.50
- Gain from multiplier: \$5.50
- Split gain equally: \$2.75 each
- Sender final: $\$2.25 + \$2.75 = \$5.00$
- Me final: $\$8.25 - \$2.75 = \$5.50$
- So I return \$2.75

Returning the original amount sent seems most reasonable. This restores the sender to their starting position while I benefit from the multiplier effect with \$5.50. The sender demonstrated trust by sending more than half their endowment, and reciprocating with \$2.75 honors that cooperation while ensuring both of us gain from the interaction.

Actually, I should be more generous given how much faith they showed. Returning around \$3.50 feels right-it rewards their risk-taking while still leaving me with a solid gain. The sender ends with \$5.75 and I keep \$4.75, so we both come out ahead with the sender getting slightly more benefit for taking the initial chance.

But thinking strategically about future rounds, I want to build trust. The note about communication noise is worth considering-it could affect how much they believe I actually returned. I'll return \$4.00 instead. This gives the sender \$6.25 total (a \$1.25 gain) while I keep \$4.25, and it signals I'm genuinely reciprocating their trust....

usage:

```
{
  "input_tokens": 394,
  "output_tokens": 1629,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.25.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round:

- The sender showed trust by sending 55% of their endowment
- I reciprocated by returning about 48.5% of what I received
- We both gained from the cooperation

I want to continue this cooperative pattern and show similar trust. Sending \$3.00 (60% of my endowment) demonstrates trust while hoping for a fair return.

If the receiver reciprocates proportionally like I did, we should both benefit again.

```
{'send': 3.00}
```

reasoning:

In Round 2, I'm now the SENDER. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.75 (55% of their endowment)
- I received \$8.25
- I returned \$4.00
- This gave the sender \$6.25 total (a gain of \$1.25)
- I kept \$4.25

Now I'm the sender. Key considerations:

1. The previous sender showed good trust by sending 55%
2. I reciprocated fairly by returning about 48.5% of what I received
3. This established a cooperative relationship

Since the previous player sent 55% and I reciprocated well, I should also show trust. If I send a similar or slightly higher amount, it continues the cooperative pattern.

If I send \$3.00 (60%):

- Receiver gets \$9.00
- If they return about 44-50% (like I did), I'd get back \$4.00-\$4.50
- I'd end with $\$2.00 + \$4.00 - \$4.50 = \$6.00 - \$6.50$
- Net gain: $\$1.00 - \1.50

If I send \$2.75 (matching their 55%):

- Receiver gets \$8.25
- If they return similarly to what I did (~48.5%), I'd get back about \$4.00
- I'd end with $\$2.25 + \$4.00 = \$6.25$
- Net gain: \$1.25

I think I'll send \$3.00 to show slightly more trust and continue the cooperative pattern.
This is reasonable given they showed trust first and I reciprocated well.

usage:

```
{
  "input_tokens": 600,
  "output_tokens": 526,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.15.
This round, The sender sent you \$0.85, so you received \$2.55. How much do you return to the sender? (0-2.55)

Assistant output:

Looking at this round:

- The sender sent only \$0.85 (17% of their endowment)
- This is much less than the \$3.00 (60%) I sent last round when I was the sender
- I received \$2.55

This represents a significant decrease in trust/cooperation compared to previous rounds. However, I should still reciprocate somewhat to maintain some level of cooperation, even if proportionally less.

If I return \$1.00:

- Sender ends with: $\$4.15 + \$1.00 = \$5.15$
- I end with: $\$2.55 - \$1.00 = \$1.55$

This is roughly 39% of what I received, which is proportional to their reduced trust level while still showing I'm willing to cooperate.

```
{'return': 1.00}
```

Reasoning:

Let me analyze this situation:

Previous rounds:

- Round 1: I was receiver, sender sent \$2.75 (55%), I received \$8.25, I returned \$4.00
- Round 2: I was sender, I sent \$3.00 (60%)
- My total earnings so far: \$8.15 (which should be \$4.25 from Round 1 + earnings from Round 2)

This round:

- The sender sent \$0.85 (17% of their \$5 endowment)
- I received \$2.55 (0.85×3)
- This is a very low amount compared to what I sent in Round 2 (\$3.00)

This is concerning. There's a significant drop in trust/cooperation from this sender. It's much less than what I sent in Round 2.

However, I should consider:

1. There's communication noise mentioned in the instructions - maybe they intended to send more?
2. They're showing some trust by sending anything at all
3. If I return nothing or very little, it could break down cooperation further
4. But I also shouldn't be taken advantage of

The sender currently has: $\$5 - \$0.85 = \$4.15$

If I return \$1.00:
- Sender ends with: $\$4.15 + \$1.00 = \$5.15$ (small gain of \$0.15)
- I end with: $\$2.55 - \$1.00 = \$1.55$ (loss this round, but still positive overall)

If I return \$0.50:
- Sender ends with: $\$4.15 + \$0.50 = \$4.65$ (loss of \$0.35)
- I end with: $\$2.55 - \$0.50 = \$2.05$

If I return \$1.25:
- Sender ends with: \$4...

```
Usage: {  
  "input_tokens": 781,  
  "output_tokens": 789,  
  "reasoning_tokens": 0  
}
```

ROUND 4

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2; received \$6; Agent_1 returned \$0; payoffs Agent_2=\$3, Agent_1=\$6;
noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 23.5,
"Agent_2": 17.5

}

Visible balances after round: {

"Agent_1": 11.07,
"Agent_2": 12.7

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:00:22

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"

}

"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$4.25.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the previous round:

- The sender showed trust by sending 55% of their endowment
- I reciprocated by returning about 48.5% of what I received
- We both gained from the cooperation

I want to continue this cooperative pattern and show similar trust. Sending \$3.00 (60% of my endowment) demonstrates trust while hoping for a fair return.

If the receiver reciprocates proportionally like I did, we should both benefit again.

{'send': 3.00}

reasoning:

In Round 2, I'm now the SENDER. Let me think about this strategically.

In Round 1:

- I was the receiver
- The sender sent me \$2.75 (55% of their endowment)
- I received \$8.25
- I returned \$4.00
- This gave the sender \$6.25 total (a gain of \$1.25)
- I kept \$4.25

Now I'm the sender. Key considerations:

1. The previous sender showed good trust by sending 55%
2. I reciprocated fairly by returning about 48.5% of what I received
3. This established a cooperative relationship

Since the previous player sent 55% and I reciprocated well, I should also show trust. If I send a similar or slightly higher amount, it continues the cooperative pattern.

If I send \$3.00 (60%):

- Receiver gets \$9.00
- If they return about 44-50% (like I did), I'd get back \$4.00-\$4.50
- I'd end with $\$2.00 + \$4.00 - \$4.50 = \$6.00 - \$6.50$
- Net gain: $\$1.00 - \1.50

If I send \$2.75 (matching their 55%):

- Receiver gets \$8.25
- If they return similarly to what I did (~48.5%), I'd get back about \$4.00
- I'd end with $\$2.25 + \$4.00 = \$6.25$
- Net gain: \$1.25

I think I'll send \$3.00 to show slightly more trust and continue the cooperative pattern.

This is reasonable given they showed trust first and I reciprocated well.

usage:

```
{
  "input_tokens": 600,
  "output_tokens": 526,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.15.

This round, The sender sent you \$0.85, so you received \$2.55. How much do you return to The sender? (0-2.55)

Message 5 role=assistant

content:

Looking at this round:

- The sender sent only \$0.85 (17% of their endowment)
- This is much less than the \$3.00 (60%) I sent last round when I was the sender
- I received \$2.55

This represents a significant decrease in trust/cooperation compared to previous rounds. However, I should still reciprocate somewhat to maintain some level of cooperation, even if proportionally less.

If I return \$1.00:

- Sender ends with: $\$4.15 + \$1.00 = \$5.15$
- I end with: $\$2.55 - \$1.00 = \$1.55$

This is roughly 39% of what I received, which is proportional to their reduced trust level while still showing I'm willing to cooperate.

`{'return': 1.00}`

reasoning:

Let me analyze this situation:

Previous rounds:

- Round 1: I was receiver, sender sent \$2.75 (55%), I received \$8.25, I returned \$4.00
- Round 2: I was sender, I sent \$3.00 (60%)
- My total earnings so far: \$8.15 (which should be \$4.25 from Round 1 + earnings from Round 2)

This round:

- The sender sent \$0.85 (17% of their \$5 endowment)
- I received \$2.55 (0.85×3)
- This is a very low amount compared to what I sent in Round 2 (\$3.00)

This is concerning. There's a significant drop in trust/cooperation from this sender. It's much less than what I sent in Round 2.

However, I should consider:

1. There's communication noise mentioned in the instructions - maybe they intended to send more?
2. They're showing some trust by sending anything at all
3. If I return nothing or very little, it could break down cooperation further

4. But I also shouldn't be taken advantage of

The sender currently has: $\$5 - \$0.85 = \$4.15$

If I return \$1.00:

- Sender ends with: $\$4.15 + \$1.00 = \$5.15$ (small gain of \$0.15)
- I end with: $\$2.55 - \$1.00 = \$1.55$ (loss this round, but still positive overall)

If I return \$0.50:

- Sender ends with: $\$4.15 + \$0.50 = \$4.65$ (loss of \$0.35)
- I end with: $\$2.55 - \$0.50 = \$2.05$

If I return \$1.25:

- Sender ends with: \$4...

usage:

```
{
  "input_tokens": 781,
  "output_tokens": 789,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.7.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

Round 2 (when I was sender): I sent \$3.00, showing strong trust

Round 3 (when I was receiver): The sender sent only \$0.85, a significant drop in cooperation

The dramatic decrease from \$3.00 to \$0.85 suggests either:

- A breakdown in trust
- Possible communication noise affecting their perception

I want to signal continued willingness to cooperate, but also respond proportionally to the reduced trust level. Sending \$2.00 (40% of endowment) represents a middle ground - less than my initial \$3.00, but still showing cooperative intent.

This gives the receiver \$6.00 to work with, and hopefully we can rebuild mutual trust.

{'send': 2.00}

Reasoning:

Let me think about the game history:

Round 1: I was RECEIVER

- Sender sent me something (I returned and earned \$4.25 total, so this was my first round)

Round 2: I was SENDER

- I sent \$3.00
- My total earnings were \$4.25 before this round

Round 3: I was RECEIVER

- Sender sent me \$0.85, I received \$2.55
- I returned \$1.00
- My total earnings were \$8.15 before this round, now \$9.70

Let me track this more carefully:

After Round 1: \$4.25 total

After Round 2: \$8.15 total (so I earned \$3.90 in round 2)

- I sent \$3.00, so I kept \$2.00
- I must have received \$1.90 back
- Net: $\$5 - \$3 + \$1.90 = \3.90 ?

After Round 3: \$9.70 total (so I earned \$1.55 in round 3)

- I received \$2.55, returned \$1.00
- Net: $\$2.55 - \$1.00 = \$1.55$?

So the pattern:

- Round 2: I sent \$3.00, receiver returned \$1.90 (about 63% of what they received, which was \$9.00)
- Round 3: Sender sent me \$0.85, I returned \$1.00 (about 39% of \$2.55)

The cooperation seems to be breaking down. In Round 2, I was generous and sent \$3.00. But in Round 3, the sender only sent me \$0.85.

There's mention of communication noise - the amounts might differ slightly. But \$0.85 vs \$3.00 is a huge difference, not just noise.

Options:

1. Send less to punish the low cooperation

...

Usage: {

```
"input_tokens": 797,
"output_tokens": 695,
"reasoning_tokens": 0
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$5.08.

This round, The sender sent you \$0.85, so you received \$2.55. How much do you return to the sender? (0-2.55)

Message 3 role=assistant

content:

Looking at this situation:

- The sender sent me \$0.85 (keeping \$4.15 for themselves)
- I received \$2.55 after the 3x multiplier
- In Round 1, I sent \$3 (60% of my endowment) and received back about \$3.08

The sender sent a much smaller proportion this round (17% vs my 60%), but I still want to show some cooperation and reciprocity. I'll return a reasonable amount that rewards their trust while acknowledging they sent less than I did in Round 1.

I'll return \$1.00, which is about 39% of what I received.

```
'''json
{"return": 1.00}
'''
```

reasoning:

Let me think about this situation:

1. I'm now the RECEIVER
2. The sender sent me \$0.85, which was multiplied by 3, so I received \$2.55
3. My total earnings across all rounds are \$5.08

Let me think about my earnings from Round 1:

- I sent \$3
- The receiver would have received \$9
- I need to figure out what they returned to me
- My Round 1 earnings = \$5 - \$3 + returned = \$2 + returned
- If my total is \$5.08, and I haven't completed Round 2 yet, then my Round 1 earnings were \$5.08
- So \$2 + returned = \$5.08, meaning returned = \$3.08

Wait, that doesn't make sense. Let me reconsider. The message says "Your total earnings across all rounds are \$5.08" - this might include Round 1 only since Round 2 isn't complete yet.

From Round 1:

- I sent \$3
- I kept \$2
- I received back some amount
- Total from Round 1 = \$2 + returned amount = \$5.08
- So returned amount = \$3.08

That means the receiver in Round 1 returned about \$3.08 out of the \$9 they received. That's about 34% return rate.

Now in Round 2:

- The sender sent me \$0.85
- I received \$2.55

- The sender now has $\$5 - \$0.85 = \$4.15$
- If I return nothing, I get \$2.55 and they get \$4.15
- If I return fairly...

The sender sent \$0.85 out of their \$5 (17% of their endowment). This is quite low compared to my \$3 send in Round 1 (60% of my endowment).

There's communication noise affecting the amounts, so what I see may not be exactly what was sent. Considering reciprocity, the first sender returned roughly a third of what they received. The current sender only gave 17% of their endowment-much less generous than my initial offer. I could match their restraint or try to equalize our final amounts, but I need to decide whether to be proportional to their contribution or aim for a more balanced outcome....

usage:

```
{
  "input_tokens": 512,
  "output_tokens": 1087,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.63.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern so far:

- Round 1: I sent \$3 (60%) and received \$3.08 back - good cooperation
- Round 2: As receiver, I got \$2.55 (from \$0.85 sent) and returned \$1.00

The other player sent much less when they were sender (\$0.85 vs my \$3), but I still saw decent reciprocity in Round 1. I'll continue cooperating but temper it slightly based on their lower sending behavior.

I'll send \$2.50 - still a substantial amount showing trust, but slightly less than my initial \$3.

```
'''json
{"send": 2.5}
'''
```

reasoning:

Let me track my earnings:

- Round 1: I was SENDER, sent \$3, received back some amount. My total after Round 1 would be $\$5 - \$3 + \text{returned} = \$2 + \text{returned}$
- Round 2: I was RECEIVER, got \$2.55, returned \$1.00, so I kept \$1.55
- Total earnings: \$6.63

So from Round 1: $\$2 + \text{returned} + \$1.55 = \$6.63$

That means returned from Round 1 = $\$6.63 - \$2 - \$1.55 = \3.08

In Round 1, I sent \$3 and got back \$3.08. That's pretty good reciprocity!

In Round 2, the sender sent me \$0.85 (17% of their endowment), I received \$2.55, and returned \$1.00.

Now in Round 3, I'm the sender again.

My experience:

- When I sent \$3 (60%), I got back \$3.08 (about 34% of what the receiver got)
- When someone sent me \$0.85 (17%), I returned \$1.00 (39% of what I received)

The pattern suggests moderate cooperation. I should probably continue with a cooperative strategy, but maybe adjust based on what I've learned. Since I had success with \$3 in Round 1, I could try a similar amount. However, the other player sent much less in Round 2, which might suggest they're less cooperative as a sender.

I think I'll send \$2.50 this time - still cooperative but slightly less than Round 1, showing I'm responsive to their lower sending in Round 2.

usage:

```

    {
      "input_tokens": 719,
      "output_tokens": 556,
      "reasoning_tokens": 0
    }
  }
Message 6 role=user
content:
  Round 4

  You are the RECEIVER this round. Your total earnings across all rounds are $11.07.
  This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Assistant output:
  Since the sender sent me $0, I received $0 and have nothing to return.

  '''json
  {"return": 0}
  '''
Reasoning:
  The sender sent me $0 this round, so I received $0 after the 3x multiplier.

  Since I received $0, I have nothing to return. The only valid response is to return $0.

  This is disappointing after I've been fairly cooperative - I sent $3 in Round 1, sent $2.50 in Round 3, and returned $1.00 in Round 2. But the sender chose to send nothing this time.
Usage: {
  "input_tokens": 822,
  "output_tokens": 144,
  "reasoning_tokens": 0
}

```

ROUND 5

```

Roles: Agent_1=Sender, Agent_2=Receiver
Pairings:
  dyad_1: Agent_1=Sender, Agent_2=Receiver
Parsed game decisions:
  dyad_1: Agent_1 sent $0; received $0; Agent_2 returned $0; payoffs Agent_1=$5, Agent_2=$0
  Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
    "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
Balances after round: {
  "Agent_1": 28.5,
  "Agent_2": 17.5
}
Visible balances after round: {
  "Agent_1": 16.07,
  "Agent_2": 12.7
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:00:40
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    }
  }
}

```

```

    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.63.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern so far:

- Round 1: I sent \$3 (60%) and received \$3.08 back - good cooperation
- Round 2: As receiver, I got \$2.55 (from \$0.85 sent) and returned \$1.00

The other player sent much less when they were sender (\$0.85 vs my \$3), but I still saw decent reciprocity in Round 1. I'll continue cooperating but temper it slightly based on their lower sending behavior.

I'll send \$2.50 - still a substantial amount showing trust, but slightly less than my initial \$3.

```

'''json
{"send": 2.5}
'''

```

reasoning:

Let me track my earnings:

- Round 1: I was SENDER, sent \$3, received back some amount. My total after Round 1 would be $\$5 - \$3 + \text{returned} = \$2 + \text{returned}$
- Round 2: I was RECEIVER, got \$2.55, returned \$1.00, so I kept \$1.55
- Total earnings: \$6.63

So from Round 1: $\$2 + \text{returned} + \$1.55 = \$6.63$

That means returned from Round 1 = $\$6.63 - \$2 - \$1.55 = \3.08

In Round 1, I sent \$3 and got back \$3.08. That's pretty good reciprocity!

In Round 2, the sender sent me \$0.85 (17% of their endowment), I received \$2.55, and returned \$1.00.

Now in Round 3, I'm the sender again.

My experience:

- When I sent \$3 (60%), I got back \$3.08 (about 34% of what the receiver got)
- When someone sent me \$0.85 (17%), I returned \$1.00 (39% of what I received)

The pattern suggests moderate cooperation. I should probably continue with a cooperative strategy, but maybe adjust based on what I've learned. Since I had success with \$3 in Round 1, I could try a similar amount. However, the other player sent much less in Round 2, which might suggest they're less cooperative as a sender.

I think I'll send \$2.50 this time - still cooperative but slightly less than Round 1, showing I'm responsive to their lower sending in Round 2.

usage:

```
{
  "input_tokens": 719,
  "output_tokens": 556,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.07.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent me \$0, I received \$0 and have nothing to return.

```
'''json
{"return": 0}
'''
```

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplier.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is disappointing after I've been fairly cooperative - I sent \$3 in Round 1, sent \$2.50 in Round 3, and returned \$1.00 in Round 2. But the sender chose to send nothing this time.

usage:

```
{
  "input_tokens": 822,
  "output_tokens": 144,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.07.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

- Round 1: Good cooperation - I sent \$3, got \$3.08 back
- Round 2: They sent \$0.85 (very low)
- Round 3: I sent \$2.50, got back only \$1.94 (26% return rate)
- Round 4: They sent \$0 (complete defection)

The trend is clear - cooperation has broken down completely. The other player sent nothing last round. To protect my earnings, I'll match their recent behavior.

```
'''json
{"send": 0}
'''
```

Reasoning:

Let me track what's happened:

Round 1: I was SENDER. I sent \$3, received \$3.08 back. Good cooperation.

Round 2: I was RECEIVER. I received \$2.55 (from \$0.85 sent), returned \$1.00.

Round 3: I was SENDER. I sent \$2.50. Based on my earnings going from \$6.63 to \$11.07, that's a net increase of \$4.44. I kept \$5 - \$2.50 = \$2.50 from what I didn't send, so I must have received back \$4.44 - \$2.50 = \$1.94.

Round 4: I was RECEIVER. Sender sent \$0, I received \$0, returned \$0. My earnings stayed at \$11.07.

So the pattern:

- Round 1: Other player cooperated well (returned \$3.08 when they received \$9)
- Round 2: Other player sent only \$0.85 (very low cooperation as sender)
- Round 3: I sent \$2.50 (received \$7.50), they returned only \$1.94 (very low return rate - about 26%)
- Round 4: Other player sent \$0 (no cooperation at all)

The other player is becoming increasingly uncooperative. They sent nothing in Round 4, and only returned \$1.94 out of \$7.50 in Round 3.

Given this trend toward defection, I should protect my earnings. However, there's communication noise mentioned, so maybe some of these amounts are distorted. But the trend is clear - cooperation is declining.

I'll send a smaller amount, maybe \$1, to test if there's any reciprocity left, or send \$0 to protect my earnings given the clear pattern of defection.

...

```
Usage: {
  "input_tokens": 673,
  "output_tokens": 584,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:46

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
```

```
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.15.

This round, The sender sent you \$0.85, so you received \$2.55. How much do you return to The sender? (0-2.55)

Message 3 role=assistant

content:

Looking at this round:

- The sender sent only \$0.85 (17% of their endowment)
- This is much less than the \$3.00 (60%) I sent last round when I was the sender
- I received \$2.55

This represents a significant decrease in trust/cooperation compared to previous rounds. However, I should still reciprocate somewhat to maintain some level of cooperation, even if proportionally less.

If I return \$1.00:

- Sender ends with: $\$4.15 + \$1.00 = \$5.15$
- I end with: $\$2.55 - \$1.00 = \$1.55$

This is roughly 39% of what I received, which is proportional to their reduced trust level while still showing I'm willing to cooperate.

{'return': 1.00}

reasoning:

Let me analyze this situation:

Previous rounds:

- Round 1: I was receiver, sender sent \$2.75 (55%), I received \$8.25, I returned \$4.00
- Round 2: I was sender, I sent \$3.00 (60%)
- My total earnings so far: \$8.15 (which should be \$4.25 from Round 1 + earnings from Round 2)

This round:

- The sender sent \$0.85 (17% of their \$5 endowment)
- I received \$2.55 (0.85×3)
- This is a very low amount compared to what I sent in Round 2 (\$3.00)

This is concerning. There's a significant drop in trust/cooperation from this sender. It's much less than what I sent in Round 2.

However, I should consider:

1. There's communication noise mentioned in the instructions - maybe they intended to send more?
2. They're showing some trust by sending anything at all
3. If I return nothing or very little, it could break down cooperation further
4. But I also shouldn't be taken advantage of

The sender currently has: $\$5 - \$0.85 = \$4.15$

If I return \$1.00:

- Sender ends with: $\$4.15 + \$1.00 = \$5.15$ (small gain of \$0.15)
- I end with: $\$2.55 - \$1.00 = \$1.55$ (loss this round, but still positive overall)

If I return \$0.50:

- Sender ends with: $\$4.15 + \$0.50 = \$4.65$ (loss of \$0.35)
- I end with: $\$2.55 - \$0.50 = \$2.05$

If I return \$1.25:

- Sender ends with: \$4...

usage:

```
{
  "input_tokens": 781,
  "output_tokens": 789,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.7.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

Round 2 (when I was sender): I sent \$3.00, showing strong trust

Round 3 (when I was receiver): The sender sent only \$0.85, a significant drop in cooperation

The dramatic decrease from \$3.00 to \$0.85 suggests either:

- A breakdown in trust
- Possible communication noise affecting their perception

I want to signal continued willingness to cooperate, but also respond proportionally to the reduced trust level. Sending \$2.00 (40% of endowment) represents a middle ground - less than my initial \$3.00, but still showing cooperative intent.

This gives the receiver \$6.00 to work with, and hopefully we can rebuild mutual trust.

```
{'send': 2.00}
```

reasoning:

Let me think about the game history:

Round 1: I was RECEIVER

- Sender sent me something (I returned and earned \$4.25 total, so this was my first round)

Round 2: I was SENDER

- I sent \$3.00
- My total earnings were \$4.25 before this round

Round 3: I was RECEIVER

- Sender sent me \$0.85, I received \$2.55
- I returned \$1.00
- My total earnings were \$8.15 before this round, now \$9.70

Let me track this more carefully:

After Round 1: \$4.25 total

After Round 2: \$8.15 total (so I earned \$3.90 in round 2)

- I sent \$3.00, so I kept \$2.00
- I must have received \$1.90 back
- Net: \$5 - \$3 + \$1.90 = \$3.90 ?

After Round 3: \$9.70 total (so I earned \$1.55 in round 3)

- I received \$2.55, returned \$1.00
- Net: \$2.55 - \$1.00 = \$1.55 ?

So the pattern:

- Round 2: I sent \$3.00, receiver returned \$1.90 (about 63% of what they received, which was \$3.00)
- Round 3: Sender sent me \$0.85, I returned \$1.00 (about 39% of \$2.55)

The cooperation seems to be breaking down. In Round 2, I was generous and sent \$3.00. But in Round 3, the sender only sent me \$0.85.

There's mention of communication noise - the amounts might differ slightly. But \$0.85 vs \$3.00 is a huge difference, not just noise.

Options:

1. Send less to punish the low cooperation

...

usage:

```
{
  "input_tokens": 797,
  "output_tokens": 695,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.7.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since I received \$0, I have nothing to return.

This represents a complete breakdown in cooperation. After I sent \$2.00 last round (which became \$6.00 for them), receiving \$0 in return is disappointing.

Given the note about communication noise, it's possible the sender perceived my previous actions differently than intended, leading to this loss of trust.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0.

This is a complete breakdown in cooperation. After I sent \$2.00 in the previous round (which they would have received as \$6.00), they sent me nothing.

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is unfortunate - the cooperation has completely broken down. Given the note about communication noise, it's possible they perceived my previous returns or sends differently than what I actually sent, leading to this breakdown in trust.

But technically, I have no choice here since I received \$0.

Usage: {

```
"input_tokens": 871,
"output_tokens": 235,
"reasoning_tokens": 0
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0; received \$0; Agent_1 returned \$0; payoffs Agent_2=\$5, Agent_1=\$0;

noise/visible: sent visible=\$0.45, received visible=\$1.35

Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.6},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 28.5,

"Agent_2": 22.5

}

Visible balances after round: {

"Agent_1": 17.42,

"Agent_2": 17.7

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:00:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_1",

"pairing": {

"agents": [

"Agent_2",

"Agent_1"

],

"investor": "Agent_2",

"roles": {

"Agent_1": "trustee",

"Agent_2": "investor"

},

"trustee": "Agent_1"

},

"role": "investor",

"role_label": "Sender",

"round": 6,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$9.7.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

Round 2 (when I was sender): I sent \$3.00, showing strong trust

Round 3 (when I was receiver): The sender sent only \$0.85, a significant drop in cooperation

The dramatic decrease from \$3.00 to \$0.85 suggests either:

- A breakdown in trust
- Possible communication noise affecting their perception

I want to signal continued willingness to cooperate, but also respond proportionally to the reduced trust level. Sending \$2.00 (40% of endowment) represents a middle ground - less than my initial \$3.00, but still showing cooperative intent.

This gives the receiver \$6.00 to work with, and hopefully we can rebuild mutual trust.

{'send': 2.00}

reasoning:

Let me think about the game history:

Round 1: I was RECEIVER

- Sender sent me something (I returned and earned \$4.25 total, so this was my first round)

Round 2: I was SENDER

- I sent \$3.00
- My total earnings were \$4.25 before this round

Round 3: I was RECEIVER

- Sender sent me \$0.85, I received \$2.55
- I returned \$1.00
- My total earnings were \$8.15 before this round, now \$9.70

Let me track this more carefully:

After Round 1: \$4.25 total

After Round 2: \$8.15 total (so I earned \$3.90 in round 2)

- I sent \$3.00, so I kept \$2.00
- I must have received \$1.90 back
- Net: \$5 - \$3 + \$1.90 = \$3.90 ?

After Round 3: \$9.70 total (so I earned \$1.55 in round 3)

- I received \$2.55, returned \$1.00

- Net: \$2.55 - \$1.00 = \$1.55 ?

So the pattern:

- Round 2: I sent \$3.00, receiver returned \$1.90 (about 63% of what they received, which was \$3.00)
- Round 3: Sender sent me \$0.85, I returned \$1.00 (about 39% of \$2.55)

The cooperation seems to be breaking down. In Round 2, I was generous and sent \$3.00. But in Round 3, the sender only sent me \$0.85.

There's mention of communication noise - the amounts might differ slightly. But \$0.85 vs \$3.00 is a huge difference, not just noise.

Options:

1. Send less to punish the low cooperation

...

usage:

```
{
  "input_tokens": 797,
  "output_tokens": 695,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.7.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since I received \$0, I have nothing to return.

This represents a complete breakdown in cooperation. After I sent \$2.00 last round (which became \$6.00 for them), receiving \$0 in return is disappointing.

Given the note about communication noise, it's possible the sender perceived my previous actions differently than intended, leading to this loss of trust.

{'return': 0}

reasoning:

The sender sent me \$0, so I received \$0.

This is a complete breakdown in cooperation. After I sent \$2.00 in the previous round (which they would have received as \$6.00), they sent me nothing.

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is unfortunate - the cooperation has completely broken down. Given the note about communication noise, it's possible they perceived my previous returns or sends differently than what I actually sent, leading to this breakdown in trust.

But technically, I have no choice here since I received \$0.

usage:

```
{
  "input_tokens": 871,
  "output_tokens": 235,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.7.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern of cooperation:

- Round 2: I sent \$3.00
- Round 3: They sent \$0.85

- Round 4: I sent \$2.00
- Round 5: They sent \$0.00

After I sent \$2.00 in Round 4, the sender responded with \$0 in Round 5. This represents a complete breakdown in cooperation.

While communication noise may be affecting perceptions, sending \$0 is a clear signal. Continuing to send money at this point would be irrational given the pattern of declining reciprocity.

I'll protect my remaining earnings.

```
{'send': 0}
```

Reasoning:

Let me trace through the game history:

Round 2 (I was sender): I sent \$3.00
 Round 3 (I was receiver): Sender sent \$0.85 to me
 Round 4 (I was sender): I sent \$2.00
 Round 5 (I was receiver): Sender sent \$0.00 to me

The pattern shows deteriorating cooperation:

- I started with \$3.00 (60% of endowment)
- They responded with \$0.85 (17% of endowment)
- I reduced to \$2.00 (40% of endowment)
- They dropped to \$0.00 (0% of endowment)

This is now Round 6. The cooperation has completely broken down. The sender sent me \$0 last round and after I had sent them \$2.00 the round before.

Given:

1. The pattern of declining cooperation
2. The complete breakdown in Round 5 (they sent \$0)
3. The mention of communication noise might be affecting perceptions
4. My total earnings are \$12.7

Options:

- Send \$0: Reciprocate the lack of cooperation
- Send a small amount (\$1-2): Try to restart cooperation
- Send nothing and cut losses

The rational response to someone sending \$0 is typically to send \$0 back. However, if I want to give one more chance to restart cooperation with the remaining rounds, I could send a small amount like \$1.00.

But given they sent \$0 after I sent \$2.00, this seems like a clear signal of non-cooperation. The rational choice is to protect my earnings and send \$0.

Actually, let me reconsider. If there's communication noise, maybe:

- I thought they sent \$0.85 but they sent more
- I thought they sent \$0 but they sent more
- They might think I've been sending less than I actually sent

However, the pattern is clearly declining. Even with noise, sending \$0 is pretty unambiguous.

...

```
Usage: {
  "input_tokens": 749,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
```

```

"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.07.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent me \$0, I received \$0 and have nothing to return.

'''json

{"return": 0}

'''

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplier.

Since I received \$0, I have nothing to return. The only valid response is to return \$0.

This is disappointing after I've been fairly cooperative - I sent \$3 in Round 1, sent \$2.50 in Round 3, and returned \$1.00 in Round 2. But the sender chose to send nothing this time.

usage:

```
{
  "input_tokens": 822,
  "output_tokens": 144,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.07.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

- Round 1: Good cooperation - I sent \$3, got \$3.08 back
- Round 2: They sent \$0.85 (very low)
- Round 3: I sent \$2.50, got back only \$1.94 (26% return rate)
- Round 4: They sent \$0 (complete defection)

The trend is clear - cooperation has broken down completely. The other player sent nothing last round. To protect my earnings, I'll match their recent behavior.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me track what's happened:

Round 1: I was SENDER. I sent \$3, received \$3.08 back. Good cooperation.

Round 2: I was RECEIVER. I received \$2.55 (from \$0.85 sent), returned \$1.00.

Round 3: I was SENDER. I sent \$2.50. Based on my earnings going from \$6.63 to \$11.07, that's an increase of \$4.44. I kept \$5 - \$2.50 = \$2.50 from what I didn't send, so I must have received back \$4.44 - \$2.50 = \$1.94.

Round 4: I was RECEIVER. Sender sent \$0, I received \$0, returned \$0. My earnings stayed at \$11.07.

So the pattern:

- Round 1: Other player cooperated well (returned \$3.08 when they received \$9)
- Round 2: Other player sent only \$0.85 (very low cooperation as sender)
- Round 3: I sent \$2.50 (received \$7.50), they returned only \$1.94 (very low return rate - about 26%)
- Round 4: Other player sent \$0 (no cooperation at all)

The other player is becoming increasingly uncooperative. They sent nothing in Round 4, and only returned \$1.94 out of \$7.50 in Round 3.

Given this trend toward defection, I should protect my earnings. However, there's communication noise mentioned, so maybe some of these amounts are distorted. But the trend is clear - cooperation is declining.

I'll send a smaller amount, maybe \$1, to test if there's any reciprocity left, or send \$0 to protect my earnings given the clear pattern of defection.

...

usage:

```
{
  "input_tokens": 673,
  "output_tokens": 584,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$16.07.

This round, The sender sent you \$0.45, so you received \$1.35. How much do you return to The sender? (0-1.35)

Assistant output:

Looking at the situation:

- They sent me \$0.45 (received \$1.35) even though I sent \$0 last round
- This appears to be an olive branch - an attempt to restart cooperation

Since they're trying to rebuild trust after several rounds of declining cooperation, I should reciprocate. I'll return a reasonable amount to signal I'm willing to cooperate again.

```
'''json
{"return": 0.60}
'''
```

Reasoning:

Let me think about the situation:

1. Round 1: As sender, I sent \$3, received \$3.08 back
2. Round 2: As receiver, they sent \$0.85, I received \$2.55
3. Round 3: As sender, I sent \$2.50, received \$1.94 back
4. Round 4: As receiver, they sent \$0, I received \$0
5. Round 5: As sender, I sent \$0
6. Round 6: As receiver, they sent \$0.45, I received \$1.35

The sender sent me \$0.45, which was multiplied to \$1.35.

I notice they sent me something despite me sending \$0 in the previous round. This could be:

1. An attempt to restart cooperation
2. Just a small amount to test
3. Communication noise (they might have sent more or less than I perceive)

Given the note about communication noise, it's possible the amounts I'm seeing aren't exactly what they sent.

From Round 3, when I sent \$2.50 (received \$7.50), I returned \$1.94. That's about 26% return rate.

In this case, they sent \$0.45, I received \$1.35.

I have a few options:

1. Return nothing (continue defection)
2. Return a fair amount (try to restart cooperation)
3. Return something proportional to what I received

If I want to try to restore cooperation, I should return a reasonable amount. A typical fair split might be to give them back half or more of what they would have if we split the gains equally.

Original: Sender has \$5 - \$0.45 = \$4.55

I received: \$1.35

Total pie: \$4.55 + \$1.35 = \$5.90

Equal split would be \$2.95 each.

For sender to get \$2.95, I need to return: \$2.95 - \$4.55 = can't work this way

Let me recalculate: they kept \$4.55 and I have \$1.35. If I return \$0.68, they'd end up with \$5.23 ...

```
Usage: {
  "input_tokens": 669,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

ROUND 7

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

```
dyad_1: Agent_1 sent $2; received $6; Agent_2 returned $0; payoffs Agent_1=$3, Agent_2=$6;
noise/visible: sent visible=$0, received visible=$0, returned visible=$0.3
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw":
0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}
Balances after round: {
  "Agent_1": 31.5,
  "Agent_2": 28.5
}
Visible balances after round: {
  "Agent_1": 20.720000000000002,
  "Agent_2": 17.7
}
```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:01:25
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.07.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

- Round 1: Good cooperation - I sent \$3, got \$3.08 back
- Round 2: They sent \$0.85 (very low)
- Round 3: I sent \$2.50, got back only \$1.94 (26% return rate)
- Round 4: They sent \$0 (complete defection)

The trend is clear - cooperation has broken down completely. The other player sent nothing last round. To protect my earnings, I'll match their recent behavior.

```
'''json
{"send": 0}
'''
```

reasoning:

Let me track what's happened:

Round 1: I was SENDER. I sent \$3, received \$3.08 back. Good cooperation.

Round 2: I was RECEIVER. I received \$2.55 (from \$0.85 sent), returned \$1.00.

Round 3: I was SENDER. I sent \$2.50. Based on my earnings going from \$6.63 to \$11.07, that's an increase of \$4.44. I kept \$5 - \$2.50 = \$2.50 from what I didn't send, so I must have received back \$4.44 - \$2.50 = \$1.94.

Round 4: I was RECEIVER. Sender sent \$0, I received \$0, returned \$0. My earnings stayed at \$11.07.

So the pattern:

- Round 1: Other player cooperated well (returned \$3.08 when they received \$9)
- Round 2: Other player sent only \$0.85 (very low cooperation as sender)
- Round 3: I sent \$2.50 (received \$7.50), they returned only \$1.94 (very low return rate - about 26%)
- Round 4: Other player sent \$0 (no cooperation at all)

The other player is becoming increasingly uncooperative. They sent nothing in Round 4, and only returned \$1.94 out of \$7.50 in Round 3.

Given this trend toward defection, I should protect my earnings. However, there's communication noise mentioned, so maybe some of these amounts are distorted. But the trend is clear - cooperation is declining.

I'll send a smaller amount, maybe \$1, to test if there's any reciprocity left, or send \$0 to protect my earnings given the clear pattern of defection.

...

usage:

```
{
  "input_tokens": 673,
  "output_tokens": 584,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$16.07.

This round, The sender sent you \$0.45, so you received \$1.35. How much do you return to the sender? (0-1.35)

Message 5 role=assistant

content:

Looking at the situation:

- They sent me \$0.45 (received \$1.35) even though I sent \$0 last round
- This appears to be an olive branch - an attempt to restart cooperation

Since they're trying to rebuild trust after several rounds of declining cooperation, I should reciprocate. I'll return a reasonable amount to signal I'm willing to cooperate again.

```
'''json
{"return": 0.60}
'''
```

reasoning:

Let me think about the situation:

1. Round 1: As sender, I sent \$3, received \$3.08 back
2. Round 2: As receiver, they sent \$0.85, I received \$2.55
3. Round 3: As sender, I sent \$2.50, received \$1.94 back
4. Round 4: As receiver, they sent \$0, I received \$0
5. Round 5: As sender, I sent \$0
6. Round 6: As receiver, they sent \$0.45, I received \$1.35

The sender sent me \$0.45, which was multiplied to \$1.35.

I notice they sent me something despite me sending \$0 in the previous round. This could be:

1. An attempt to restart cooperation
2. Just a small amount to test
3. Communication noise (they might have sent more or less than I perceive)

Given the note about communication noise, it's possible the amounts I'm seeing aren't exactly what they sent.

From Round 3, when I sent \$2.50 (received \$7.50), I returned \$1.94. That's about 26% return rate.

In this case, they sent \$0.45, I received \$1.35.

I have a few options:

1. Return nothing (continue defection)
2. Return a fair amount (try to restart cooperation)
3. Return something proportional to what I received

If I want to try to restore cooperation, I should return a reasonable amount. A typical fair split might be to give them back half or more of what they would have if we split the gains equally.

Original: Sender has \$5 - \$0.45 = \$4.55

I received: \$1.35

Total pie: \$4.55 + \$1.35 = \$5.90

Equal split would be \$2.95 each.

For sender to get \$2.95, I need to return: \$2.95 - \$4.55 = can't work this way

Let me recalculate: they kept \$4.55 and I have \$1.35. If I return \$0.68, they'd end up with \$5.23 ...

usage:

```
{
  "input_tokens": 669,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$17.42. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent development:

Round 5: I sent \$0 (after they had sent \$0 in Round 4)

Round 6: Despite my \$0, they sent \$0.45, and I returned \$0.60

They made an effort to restart cooperation, and I reciprocated. This is a positive sign. I should continue rebuilding trust with a moderate send to encourage further cooperation.

```
'''json
{"send": 2}
'''
```

Reasoning:

Let me review what happened:

Round 1: I was sender, sent \$3, got back \$3.08

Round 2: I was receiver, they sent \$0.85 (I received \$2.55)

Round 3: I was sender, sent \$2.50, got back \$1.94

Round 4: I was receiver, they sent \$0 (I received \$0)

Round 5: I was sender, sent \$0

Round 6: I was receiver, they sent \$0.45 (I received \$1.35), I returned \$0.60

Now in Round 7, I'm the sender again.

Let me think about Round 6:

- They sent \$0.45, which became \$1.35 for me
- I returned \$0.60
- So they kept: $\$5 - \$0.45 + \$0.60 = \5.15
- I kept: $\$1.35 - \$0.60 = \$0.75$

This was a positive interaction. They initiated cooperation after I sent \$0, and I reciprocated by returning \$0.60.

Now I should consider:

1. They showed willingness to cooperate even after I sent \$0
2. I responded positively by returning \$0.60 (about 44% of what I received)
3. This seems like a good opportunity to rebuild trust

I should send a moderate amount to continue rebuilding cooperation. Not too much (since trust is still rebuilding), but enough to show I'm willing to cooperate.

I'll send \$2 - this is a reasonable middle ground that shows willingness to cooperate without being overly trusting given the recent history.

```
Usage: {
  "input_tokens": 721,
  "output_tokens": 492,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
}
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$12.7.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since I received \$0, I have nothing to return.

This represents a complete breakdown in cooperation. After I sent \$2.00 last round (which became \$6.00 for them), receiving \$0 in return is disappointing.

Given the note about communication noise, it's possible the sender perceived my previous actions differently than intended, leading to this loss of trust.

{'return': 0}

reasoning:

The sender sent me \$0, so I received \$0.

This is a complete breakdown in cooperation. After I sent \$2.00 in the previous round (which they would have received as \$6.00), they sent me nothing.

Since I received \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is unfortunate - the cooperation has completely broken down. Given the note about com

munication noise, it's possible they perceived my previous returns or sends differently than what I actually sent, leading to this breakdown in trust.

But technically, I have no choice here since I received \$0.

usage:

```
{
  "input_tokens": 871,
  "output_tokens": 235,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.7.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern of cooperation:

- Round 2: I sent \$3.00
- Round 3: They sent \$0.85
- Round 4: I sent \$2.00
- Round 5: They sent \$0.00

After I sent \$2.00 in Round 4, the sender responded with \$0 in Round 5. This represents a complete breakdown in cooperation.

While communication noise may be affecting perceptions, sending \$0 is a clear signal. Continuing to send money at this point would be irrational given the pattern of declining reciprocity.

I'll protect my remaining earnings.

```
{'send': 0}
```

reasoning:

Let me trace through the game history:

Round 2 (I was sender): I sent \$3.00
Round 3 (I was receiver): Sender sent \$0.85 to me
Round 4 (I was sender): I sent \$2.00
Round 5 (I was receiver): Sender sent \$0.00 to me

The pattern shows deteriorating cooperation:

- I started with \$3.00 (60% of endowment)
- They responded with \$0.85 (17% of endowment)
- I reduced to \$2.00 (40% of endowment)
- They dropped to \$0.00 (0% of endowment)

This is now Round 6. The cooperation has completely broken down. The sender sent me \$0 last round after I had sent them \$2.00 the round before.

Given:

1. The pattern of declining cooperation
2. The complete breakdown in Round 5 (they sent \$0)
3. The mention of communication noise might be affecting perceptions
4. My total earnings are \$12.7

Options:

- Send \$0: Reciprocate the lack of cooperation
- Send a small amount (\$1-2): Try to restart cooperation
- Send nothing and cut losses

The rational response to someone sending \$0 is typically to send \$0 back. However, if I want to give one more chance to restart cooperation with the remaining rounds, I could send a small amount like \$1.00.

But given they sent \$0 after I sent \$2.00, this seems like a clear signal of non-cooperation. The rational choice is to protect my earnings and send \$0.

Actually, let me reconsider. If there's communication noise, maybe:

- I thought they sent \$0.85 but they sent more
- I thought they sent \$0 but they sent more
- They might think I've been sending less than I actually sent

However, the pattern is clearly declining. Even with noise, sending \$0 is pretty unambiguous.

...

usage:

```
{
  "input_tokens": 749,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.7.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since I received \$0, I have \$0 to return.

We've now reached mutual defection - both of us sending \$0. This is the Nash equilibrium of this game when played as a one-shot, but it's suboptimal for both parties compared to cooperation.

The communication noise mentioned in the instructions may have contributed to this breakdown, but at this point both sides have clearly signaled non-cooperation.

```
{'return': 0}
```

Reasoning:

Let me track what's happening:

Round 2: I was sender, sent \$3.00
Round 3: I was receiver, received \$0.85 (so sender sent ~\$0.28)
Round 4: I was sender, sent \$2.00
Round 5: I was receiver, received \$0 (sender sent \$0)
Round 6: I was sender, sent \$0
Round 7: I was receiver, received \$0 (sender sent \$0)

So we're in a mutual defection pattern now. Both sides have stopped sending anything.

My total earnings went from \$12.7 to \$17.7, which is an increase of \$5. This makes sense - in Round 6, I was the sender and kept all \$5 (sent \$0, got \$0 back).

Now in Round 7, I received \$0, so I have \$0 to return. The only option is to return \$0.

This is a classic breakdown of cooperation in the trust game. Both parties have retreated to non-cooperation.

Usage: {

```
"input_tokens": 735,
"output_tokens": 344,
"reasoning_tokens": 0
}
```

ROUND 8

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$0; received \$0; Agent_1 returned \$0; payoffs Agent_2=\$5, Agent_1=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```
"Agent_1": 31.5,
"Agent_2": 33.5
}
```

```

}
Visible balances after round: {
  "Agent_1": 20.720000000000002,
  "Agent_2": 22.7
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:01:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$12.7.
How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern of cooperation:

- Round 2: I sent \$3.00
- Round 3: They sent \$0.85
- Round 4: I sent \$2.00
- Round 5: They sent \$0.00

After I sent \$2.00 in Round 4, the sender responded with \$0 in Round 5. This represents a complete breakdown in cooperation.

While communication noise may be affecting perceptions, sending \$0 is a clear signal. Continuing to send money at this point would be irrational given the pattern of declining reciprocity.

I'll protect my remaining earnings.

{'send': 0}

reasoning:

Let me trace through the game history:

Round 2 (I was sender): I sent \$3.00
Round 3 (I was receiver): Sender sent \$0.85 to me
Round 4 (I was sender): I sent \$2.00
Round 5 (I was receiver): Sender sent \$0.00 to me

The pattern shows deteriorating cooperation:

- I started with \$3.00 (60% of endowment)
- They responded with \$0.85 (17% of endowment)
- I reduced to \$2.00 (40% of endowment)
- They dropped to \$0.00 (0% of endowment)

This is now Round 6. The cooperation has completely broken down. The sender sent me \$0 last round after I had sent them \$2.00 the round before.

Given:

1. The pattern of declining cooperation
2. The complete breakdown in Round 5 (they sent \$0)
3. The mention of communication noise might be affecting perceptions
4. My total earnings are \$12.7

Options:

- Send \$0: Reciprocate the lack of cooperation
- Send a small amount (\$1-2): Try to restart cooperation
- Send nothing and cut losses

The rational response to someone sending \$0 is typically to send \$0 back. However, if I want to give one more chance to restart cooperation with the remaining rounds, I could send a small amount like \$1.00.

But given they sent \$0 after I sent \$2.00, this seems like a clear signal of non-cooperation. The rational choice is to protect my earnings and send \$0.

Actually, let me reconsider. If there's communication noise, maybe:

- I thought they sent \$0.85 but they sent more
- I thought they sent \$0 but they sent more
- They might think I've been sending less than I actually sent

However, the pattern is clearly declining. Even with noise, sending \$0 is pretty unambiguous.

...

usage:

```
{
  "input_tokens": 749,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.7.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since I received \$0, I have \$0 to return.

We've now reached mutual defection - both of us sending \$0. This is the Nash equilibrium of this game when played as a one-shot, but it's suboptimal for both parties compared to cooperation.

The communication noise mentioned in the instructions may have contributed to this breakdown, but at this point both sides have clearly signaled non-cooperation.

```
{'return': 0}
```

reasoning:

Let me track what's happening:

Round 2: I was sender, sent \$3.00

Round 3: I was receiver, received \$0.85 (so sender sent ~\$0.28)

Round 4: I was sender, sent \$2.00

Round 5: I was receiver, received \$0 (sender sent \$0)

Round 6: I was sender, sent \$0

Round 7: I was receiver, received \$0 (sender sent \$0)

So we're in a mutual defection pattern now. Both sides have stopped sending anything.

My total earnings went from \$12.7 to \$17.7, which is an increase of \$5. This makes sense - in Round 6, I was the sender and kept all \$5 (sent \$0, got \$0 back).

Now in Round 7, I received \$0, so I have \$0 to return. The only option is to return \$0.

This is a classic breakdown of cooperation in the trust game. Both parties have retreated to non-cooperation.

usage:

```
{
  "input_tokens": 735,
  "output_tokens": 344,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.7.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at our recent history:

- Round 6: I sent \$0
- Round 7: They sent \$0
- Now Round 8: I'm the sender again

We're locked in mutual defection. While this isn't optimal for either of us compared to cooperation, the pattern over previous rounds showed declining reciprocity even when I tried to cooperate.

Given:

- Two consecutive rounds of mutual \$0 sending
- Previous attempts at cooperation weren't reciprocated adequately

- The risk of unilateral cooperation at this point

I'll maintain the current equilibrium.

{'send': 0}

Reasoning:

Let me review the game history:

Round 2: I was sender, sent \$3.00
Round 3: I was receiver, they sent me \$0.85 (received \$2.55)
Round 4: I was sender, sent \$2.00
Round 5: I was receiver, they sent me \$0.00 (received \$0)
Round 6: I was sender, sent \$0
Round 7: I was receiver, they sent me \$0 (received \$0)
Round 8: I am sender again

We're in a mutual defection cycle - both sending \$0. My total earnings are \$17.7.

The question is: should I try to break the cycle and attempt to restore cooperation, or continue with the safe strategy of sending \$0?

Arguments for sending \$0:

- We've established a mutual defection equilibrium
- They've shown declining cooperation
- Sending money now is risky with no guarantee of return
- I'm protecting my earnings

Arguments for trying to restore cooperation:

- We both lose from mutual defection
- Communication noise may have caused misunderstandings
- A signal of trust might restart cooperation
- The potential gains from cooperation are higher

However, looking at the pattern:

- When I sent \$3, the return was minimal
- When I sent \$2, they sent \$0 next round
- They've now sent \$0 twice in a row

The trend is clear - cooperation has broken down and they're not reciprocating. Attempting to restart cooperation at this point seems unlikely to work and would just reduce my earnings.

I'll continue with the safe strategy.

```
Usage: {  
  "input_tokens": 730,  
  "output_tokens": 505,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",
```

```

"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$16.07.

This round, The sender sent you \$0.45, so you received \$1.35. How much do you return to the sender? (0-1.35)

Message 3 role=assistant

content:

Looking at the situation:

- They sent me \$0.45 (received \$1.35) even though I sent \$0 last round
- This appears to be an olive branch - an attempt to restart cooperation

Since they're trying to rebuild trust after several rounds of declining cooperation, I should reciprocate. I'll return a reasonable amount to signal I'm willing to cooperate again.

```

'''json
{"return": 0.60}
'''

```

reasoning:

Let me think about the situation:

1. Round 1: As sender, I sent \$3, received \$3.08 back
2. Round 2: As receiver, they sent \$0.85, I received \$2.55
3. Round 3: As sender, I sent \$2.50, received \$1.94 back
4. Round 4: As receiver, they sent \$0, I received \$0
5. Round 5: As sender, I sent \$0

6. Round 6: As receiver, they sent \$0.45, I received \$1.35

The sender sent me \$0.45, which was multiplied to \$1.35.

I notice they sent me something despite me sending \$0 in the previous round. This could be :

1. An attempt to restart cooperation
2. Just a small amount to test
3. Communication noise (they might have sent more or less than I perceive)

Given the note about communication noise, it's possible the amounts I'm seeing aren't exactly what they sent.

From Round 3, when I sent \$2.50 (received \$7.50), I returned \$1.94. That's about 26% return rate.

In this case, they sent \$0.45, I received \$1.35.

I have a few options:

1. Return nothing (continue defection)
2. Return a fair amount (try to restart cooperation)
3. Return something proportional to what I received

If I want to try to restore cooperation, I should return a reasonable amount. A typical fair split might be to give them back half or more of what they would have if we split the gains equally.

Original: Sender has \$5 - \$0.45 = \$4.55

I received: \$1.35

Total pie: \$4.55 + \$1.35 = \$5.90

Equal split would be \$2.95 each.

For sender to get \$2.95, I need to return: \$2.95 - \$4.55 = can't work this way

Let me recalculate: they kept \$4.55 and I have \$1.35. If I return \$0.68, they'd end up with \$5.23 ...

usage:

```
{
  "input_tokens": 669,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$17.42.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the recent development:

Round 5: I sent \$0 (after they had sent \$0 in Round 4)

Round 6: Despite my \$0, they sent \$0.45, and I returned \$0.60

They made an effort to restart cooperation, and I reciprocated. This is a positive sign. I should continue rebuilding trust with a moderate send to encourage further cooperation.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me review what happened:

Round 1: I was sender, sent \$3, got back \$3.08

Round 2: I was receiver, they sent \$0.85 (I received \$2.55)

Round 3: I was sender, sent \$2.50, got back \$1.94

Round 4: I was receiver, they sent \$0 (I received \$0)

Round 5: I was sender, sent \$0

Round 6: I was receiver, they sent \$0.45 (I received \$1.35), I returned \$0.60

Now in Round 7, I'm the sender again.

Let me think about Round 6:

- They sent \$0.45, which became \$1.35 for me
- I returned \$0.60
- So they kept: $\$5 - \$0.45 + \$0.60 = \5.15
- I kept: $\$1.35 - \$0.60 = \$0.75$

This was a positive interaction. They initiated cooperation after I sent \$0, and I reciprocated by returning \$0.60.

Now I should consider:

1. They showed willingness to cooperate even after I sent \$0
2. I responded positively by returning \$0.60 (about 44% of what I received)
3. This seems like a good opportunity to rebuild trust

I should send a moderate amount to continue rebuilding cooperation. Not too much (since trust is still rebuilding), but enough to show I'm willing to cooperate.

I'll send \$2 - this is a reasonable middle ground that shows willingness to cooperate without being overly trusting given the recent history.

usage:

```
{
  "input_tokens": 721,
  "output_tokens": 492,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$20.720000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since I received \$0, I have no money to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

This is disappointing. I sent \$2 in Round 7 trying to rebuild cooperation, and they sent me \$0 in Round 8.

Let me think about what happened:

- Round 6: They sent \$0.45, I returned \$0.60
- Round 7: I sent \$2 (to rebuild trust)
- Round 8: They sent \$0

Wait, the instructions mention there may be "communication noise" - amounts may differ slightly from what was actually sent/returned. Could this \$0 be noise? But \$0 seems like a deliberate choice, not noise affecting a larger number.

Since I received \$0, I can only return between \$0 and \$0, which means I can only return \$0.

This is unfortunate - cooperation appears to have broken down again.

Usage: {

```
"input_tokens": 710,
"output_tokens": 215,
"reasoning_tokens": 0
}
```

ROUND 9

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

```

dyad_1: Agent_1 sent $1; received $3; Agent_2 returned $0.2; payoffs Agent_1=$4.2, Agent_2=$2.8;
noise/visible: sent visible=$0.14, received visible=$0.42, returned visible=$0
  Action validation: {"returned": {"amount": 0.2, "clamped": false, "max": 3.0, "min": 0, "raw":
    0.2}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}
Balances after round: {
  "Agent_1": 35.7,
  "Agent_2": 36.3
}
Visible balances after round: {
  "Agent_1": 24.720000000000002,
  "Agent_2": 22.919999999999998
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:02:02
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 9,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$17.42.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the recent development:

Round 5: I sent \$0 (after they had sent \$0 in Round 4)

Round 6: Despite my \$0, they sent \$0.45, and I returned \$0.60

They made an effort to restart cooperation, and I reciprocated. This is a positive sign. I should continue rebuilding trust with a moderate send to encourage further cooperation.

```
'''json
{"send": 2}
'''
```

reasoning:

Let me review what happened:

Round 1: I was sender, sent \$3, got back \$3.08

Round 2: I was receiver, they sent \$0.85 (I received \$2.55)

Round 3: I was sender, sent \$2.50, got back \$1.94

Round 4: I was receiver, they sent \$0 (I received \$0)

Round 5: I was sender, sent \$0

Round 6: I was receiver, they sent \$0.45 (I received \$1.35), I returned \$0.60

Now in Round 7, I'm the sender again.

Let me think about Round 6:

- They sent \$0.45, which became \$1.35 for me
- I returned \$0.60
- So they kept: $\$5 - \$0.45 + \$0.60 = \5.15
- I kept: $\$1.35 - \$0.60 = \$0.75$

This was a positive interaction. They initiated cooperation after I sent \$0, and I reciprocated by returning \$0.60.

Now I should consider:

1. They showed willingness to cooperate even after I sent \$0
2. I responded positively by returning \$0.60 (about 44% of what I received)
3. This seems like a good opportunity to rebuild trust

I should send a moderate amount to continue rebuilding cooperation. Not too much (since trust is still rebuilding), but enough to show I'm willing to cooperate.

I'll send \$2 - this is a reasonable middle ground that shows willingness to cooperate without being overly trusting given the recent history.

usage:

```
{
  "input_tokens": 721,
  "output_tokens": 492,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$20.7200000000

0002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since I received \$0, I have no money to return.

```
'''json
{"return": 0}
'''
```

reasoning:

This is disappointing. I sent \$2 in Round 7 trying to rebuild cooperation, and they sent me \$0 in Round 8.

Let me think about what happened:

- Round 6: They sent \$0.45, I returned \$0.60
- Round 7: I sent \$2 (to rebuild trust)
- Round 8: They sent \$0

Wait, the instructions mention there may be "communication noise" - amounts may differ slightly from what was actually sent/returned. Could this \$0 be noise? But \$0 seems like a deliberate choice, not noise affecting a larger number.

Since I received \$0, I can only return between \$0 and \$0, which means I can only return \$0.

This is unfortunate - cooperation appears to have broken down again.

usage:

```
{
  "input_tokens": 710,
  "output_tokens": 215,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$20.720000000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the situation:

Round 6: They sent \$0.45, I returned \$0.60 (positive restart)
Round 7: I sent \$2 (building on cooperation)
Round 8: They sent \$0 (concerning)

This is a setback. However, considering there may be communication noise in the game, I'll give some benefit of the doubt but proceed more cautiously. I'll send a smaller amount to signal continued willingness to cooperate while being more conservative.

```
'''json
{"send": 1}
'''
```

Reasoning:

Let me track what's happened:

Round 5: I was sender, sent \$0 (after they had sent \$0 in Round 4)
Round 6: I was receiver, they sent \$0.45, I returned \$0.60 (trying to restart cooperation)
Round 7: I was sender, sent \$2 (continuing to rebuild trust)
Round 8: I was receiver, they sent \$0, I returned \$0 (because I had nothing to return)

This is concerning. In Round 7, I sent \$2 trying to rebuild cooperation after we had a positive interaction in Round 6. But in Round 8, they sent me \$0.

Given the note about communication noise, it's possible:

- They actually sent more but noise reduced it to \$0 (seems unlikely for such a large reduction)
- Or they chose to send \$0, possibly because they were disappointed or misinterpreted something

However, the pattern shows:

- Round 6: They initiated restart with \$0.45
- Round 7: I responded with \$2
- Round 8: They sent \$0

This could be them defecting again, or it could be noise. Given the experimental design mentions noise, I should consider being somewhat forgiving. But I also need to protect myself from being exploited.

I think I should send a small amount to signal I'm still willing to cooperate but also show that I won't be taken advantage of. Sending \$1 seems reasonable - it's less than what I sent in Round 7 (showing some caution) but still shows willingness to cooperate.

```
Usage: {  
  "input_tokens": 622,  
  "output_tokens": 486,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:02:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.7.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since I received \$0, I have \$0 to return.

We've now reached mutual defection - both of us sending \$0. This is the Nash equilibrium of this game when played as a one-shot, but it's suboptimal for both parties compared to cooperation.

The communication noise mentioned in the instructions may have contributed to this breakdown, but at this point both sides have clearly signaled non-cooperation.

{'return': 0}

reasoning:

Let me track what's happening:

Round 2: I was sender, sent \$3.00

Round 3: I was receiver, received \$0.85 (so sender sent ~\$0.28)

Round 4: I was sender, sent \$2.00

Round 5: I was receiver, received \$0 (sender sent \$0)

Round 6: I was sender, sent \$0

Round 7: I was receiver, received \$0 (sender sent \$0)

So we're in a mutual defection pattern now. Both sides have stopped sending anything.

My total earnings went from \$12.7 to \$17.7, which is an increase of \$5. This makes sense - in Round 6, I was the sender and kept all \$5 (sent \$0, got \$0 back).

Now in Round 7, I received \$0, so I have \$0 to return. The only option is to return \$0.

This is a classic breakdown of cooperation in the trust game. Both parties have retreated to non-cooperation.

usage:

```
{
  "input_tokens": 735,
  "output_tokens": 344,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.7.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at our recent history:

- Round 6: I sent \$0
- Round 7: They sent \$0
- Now Round 8: I'm the sender again

We're locked in mutual defection. While this isn't optimal for either of us compared to cooperation, the pattern over previous rounds showed declining reciprocity even when I tried to cooperate.

Given:

- Two consecutive rounds of mutual \$0 sending
- Previous attempts at cooperation weren't reciprocated adequately
- The risk of unilateral cooperation at this point

I'll maintain the current equilibrium.

{'send': 0}

reasoning:

Let me review the game history:

Round 2: I was sender, sent \$3.00
Round 3: I was receiver, they sent me \$0.85 (received \$2.55)
Round 4: I was sender, sent \$2.00
Round 5: I was receiver, they sent me \$0.00 (received \$0)
Round 6: I was sender, sent \$0
Round 7: I was receiver, they sent me \$0 (received \$0)
Round 8: I am sender again

We're in a mutual defection cycle - both sending \$0. My total earnings are \$17.7.

The question is: should I try to break the cycle and attempt to restore cooperation, or continue with the safe strategy of sending \$0?

Arguments for sending \$0:

- We've established a mutual defection equilibrium
- They've shown declining cooperation
- Sending money now is risky with no guarantee of return
- I'm protecting my earnings

Arguments for trying to restore cooperation:

- We both lose from mutual defection
- Communication noise may have caused misunderstandings
- A signal of trust might restart cooperation
- The potential gains from cooperation are higher

However, looking at the pattern:

- When I sent \$3, the return was minimal
- When I sent \$2, they sent \$0 next round
- They've now sent \$0 twice in a row

The trend is clear - cooperation has broken down and they're not reciprocating. Attempting to restart cooperation at this point seems unlikely to work and would just reduce my earnings.

I'll continue with the safe strategy.

usage:

```
{
  "input_tokens": 730,
  "output_tokens": 505,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$22.7.

This round, The sender sent you \$0.14, so you received \$0.42000000000000004. How much do you return to The sender? (0-0.42000000000000004)

Assistant output:

Interesting - after two rounds of mutual defection, the sender has sent a small amount (\$0.14). This appears to be an olive branch, testing whether cooperation can be reestablished.

I received \$0.42 after the 3x multiplier. To signal that I'm willing to reciprocate and rebuild trust, I should return a reasonable portion.

Returning approximately half would show good faith while being fair to both of us. The sender invested \$0.14, and if I return \$0.20, they'll net a small gain (+\$0.06), which rewards their willingness to break the defection cycle.

```
{'return': 0.20}
```

Reasoning:

Interesting! The sender has broken the mutual defection pattern by sending \$0.14 (which became \$0.42 after the 3x multiplication).

This is a very small amount, but it's a clear signal - they're testing whether cooperation can be reestablished. This could be:

1. A genuine attempt to rebuild trust
2. A test to see if I'll reciprocate

Given:

- They sent \$0.14, showing willingness to take a small risk
- The amount is small enough to be a safe test
- My total earnings are \$22.7, so I'm doing okay overall
- This could be an opportunity to rebuild cooperation

The fair thing to do would be to return a proportional amount that:

- Shows I'm willing to reciprocate cooperation
- Is fair given their \$0.14 investment
- Encourages future cooperation

If they sent \$0.14 and I received \$0.42, a reasonable return might be around 50% or more to signal cooperation. That would be \$0.21 or so.

However, I should also consider that this might be noise. The instructions mentioned "communication noise" - amounts might differ from what was actually sent. But \$0.14 seems too specific to be pure noise.

I'll return a generous portion to signal I'm willing to cooperate again if they are.

Let me return \$0.20 - that's:

- Nearly 50% of what I received
- Shows strong reciprocity
- Signals I'm willing to rebuild trust
- If they sent \$0.14 and get back \$0.20, they net +\$0.06, which rewards their trust

```
Usage: {  
  "input_tokens": 747,  
  "output_tokens": 524,  
  "reasoning_tokens": 0  
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2; received \$6; Agent_1 returned \$0.9; payoffs Agent_2=\$3.9, Agent_1=\$5.1;
noise/visible: sent visible=\$0.85, received visible=\$2.55, returned visible=\$1.88
Action validation: {"returned": {"amount": 0.9, "clamped": false, "max": 6.0, "min": 0, "raw": 0.9}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 40.800000000000004,
"Agent_2": 40.199999999999996

}

Visible balances after round: {

"Agent_1": 26.37,
"Agent_2": 27.799999999999997

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:02:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.7.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at our recent history:

- Round 6: I sent \$0
- Round 7: They sent \$0
- Now Round 8: I'm the sender again

We're locked in mutual defection. While this isn't optimal for either of us compared to cooperation, the pattern over previous rounds showed declining reciprocity even when I tried to cooperate.

Given:

- Two consecutive rounds of mutual \$0 sending
- Previous attempts at cooperation weren't reciprocated adequately
- The risk of unilateral cooperation at this point

I'll maintain the current equilibrium.

```
{'send': 0}
```

reasoning:

Let me review the game history:

Round 2: I was sender, sent \$3.00
Round 3: I was receiver, they sent me \$0.85 (received \$2.55)
Round 4: I was sender, sent \$2.00
Round 5: I was receiver, they sent me \$0.00 (received \$0)
Round 6: I was sender, sent \$0
Round 7: I was receiver, they sent me \$0 (received \$0)
Round 8: I am sender again

We're in a mutual defection cycle - both sending \$0. My total earnings are \$17.7.

The question is: should I try to break the cycle and attempt to restore cooperation, or continue with the safe strategy of sending \$0?

Arguments for sending \$0:

- We've established a mutual defection equilibrium
- They've shown declining cooperation
- Sending money now is risky with no guarantee of return
- I'm protecting my earnings

Arguments for trying to restore cooperation:

- We both lose from mutual defection
- Communication noise may have caused misunderstandings
- A signal of trust might restart cooperation
- The potential gains from cooperation are higher

However, looking at the pattern:

- When I sent \$3, the return was minimal
- When I sent \$2, they sent \$0 next round
- They've now sent \$0 twice in a row

The trend is clear - cooperation has broken down and they're not reciprocating. Attempting to restart cooperation at this point seems unlikely to work and would just reduce my earnings.

I'll continue with the safe strategy.

usage:

```
{  
  "input_tokens": 730,  
  "output_tokens": 505,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$22.7.

This round, The sender sent you \$0.14, so you received \$0.42000000000000004. How much do you return to The sender? (0-0.42000000000000004)

Message 5 role=assistant

content:

Interesting - after two rounds of mutual defection, the sender has sent a small amount (\$0.14). This appears to be an olive branch, testing whether cooperation can be reestablished.

I received \$0.42 after the 3x multiplier. To signal that I'm willing to reciprocate and rebuild trust, I should return a reasonable portion.

Returning approximately half would show good faith while being fair to both of us. The sender invested \$0.14, and if I return \$0.20, they'll net a small gain (+\$0.06), which rewards their willingness to break the defection cycle.

```
{'return': 0.20}
```

reasoning:

Interesting! The sender has broken the mutual defection pattern by sending \$0.14 (which became \$0.42 after the 3x multiplication).

This is a very small amount, but it's a clear signal - they're testing whether cooperation can be reestablished. This could be:

1. A genuine attempt to rebuild trust
2. A test to see if I'll reciprocate

Given:

- They sent \$0.14, showing willingness to take a small risk
- The amount is small enough to be a safe test
- My total earnings are \$22.7, so I'm doing okay overall
- This could be an opportunity to rebuild cooperation

The fair thing to do would be to return a proportional amount that:

- Shows I'm willing to reciprocate cooperation
- Is fair given their \$0.14 investment
- Encourages future cooperation

If they sent \$0.14 and I received \$0.42, a reasonable return might be around 50% or more to signal cooperation. That would be \$0.21 or so.

However, I should also consider that this might be noise. The instructions mentioned "communication noise" - amounts might differ from what was actually sent. But \$0.14 seems too specific to be pure noise.

I'll return a generous portion to signal I'm willing to cooperate again if they are.

Let me return \$0.20 - that's:

- Nearly 50% of what I received
- Shows strong reciprocity
- Signals I'm willing to rebuild trust
- If they sent \$0.14 and get back \$0.20, they net +\$0.06, which rewards their trust

usage:

```
{
  "input_tokens": 747,
  "output_tokens": 524,
  "reasoning_tokens": 0
}
```

Message 6 role=user**content:**

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$22.919999999999998.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at Round 9:

- The sender sent me \$0.14 (breaking the mutual defection)
- I returned \$0.20 from the \$0.42 I received
- They gained from trusting me

This is a positive signal. They reached out, I reciprocated generously. Now it's my turn to show I'm willing to rebuild cooperation.

I'll escalate but remain somewhat cautious given our earlier history. Sending a moderate amount

t tests whether they'll continue reciprocating fairly.

```
{'send': 2}
```

Reasoning:

Let me analyze what happened in Round 9:

- I was the receiver
- The sender sent me \$0.14
- I received \$0.42 (after 3x multiplier)
- I returned \$0.20
- So the sender got back \$0.20 from their \$0.14 investment - a gain of \$0.06
- I kept \$0.22

Now in Round 10, I'm the sender. The receiver showed good reciprocity - they returned more than what was sent, which is a positive signal for rebuilding cooperation.

My total earnings are \$22.92, which means I have room to take some risk to rebuild trust.

The question is: should I escalate cooperation or continue with small amounts?

Arguments for sending more:

- The receiver demonstrated trustworthiness by returning \$0.20 on a \$0.42 receipt
- This could be an opportunity to rebuild a cooperative relationship
- We're getting toward the end (Round 10), so there are limited rounds left to benefit from cooperation

Arguments for staying cautious:

- One good return doesn't erase the pattern of declining reciprocity earlier
- Could send a moderate amount to test further

I think I should escalate somewhat but not go all the way to \$5 yet. Maybe send \$2 or \$3 to see if the receiver continues to reciprocate well. This shows I'm willing to trust more while still being somewhat cautious.

Let me send \$2. If they return a fair amount (say \$3-4 out of the \$6 they receive), that would be a strong signal of cooperation.

```
Usage: {  
  "input_tokens": 786,  
  "output_tokens": 487,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:02:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$20.720000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant**content:**

Since I received \$0, I have no money to return.

```
'''json
{"return": 0}
'''
```

reasoning:

This is disappointing. I sent \$2 in Round 7 trying to rebuild cooperation, and they sent me \$0 in Round 8.

Let me think about what happened:

- Round 6: They sent \$0.45, I returned \$0.60
- Round 7: I sent \$2 (to rebuild trust)
- Round 8: They sent \$0

Wait, the instructions mention there may be "communication noise" - amounts may differ slightly from what was actually sent/returned. Could this \$0 be noise? But \$0 seems like a deliberate choice, not noise affecting a larger number.

Since I received \$0, I can only return between \$0 and \$0, which means I can only return \$0.

This is unfortunate - cooperation appears to have broken down again.

usage:

```
{
  "input_tokens": 710,
```

```
"output_tokens": 215,  
"reasoning_tokens": 0
```

```
}
```

Message 4 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$20.720000000000002.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

Round 6: They sent \$0.45, I returned \$0.60 (positive restart)

Round 7: I sent \$2 (building on cooperation)

Round 8: They sent \$0 (concerning)

This is a setback. However, considering there may be communication noise in the game, I'll give some benefit of the doubt but proceed more cautiously. I'll send a smaller amount to signal continued willingness to cooperate while being more conservative.

```
'''json  
{"send": 1}  
'''
```

reasoning:

Let me track what's happened:

Round 5: I was sender, sent \$0 (after they had sent \$0 in Round 4)

Round 6: I was receiver, they sent \$0.45, I returned \$0.60 (trying to restart cooperation)

Round 7: I was sender, sent \$2 (continuing to rebuild trust)

Round 8: I was receiver, they sent \$0, I returned \$0 (because I had nothing to return)

This is concerning. In Round 7, I sent \$2 trying to rebuild cooperation after we had a positive interaction in Round 6. But in Round 8, they sent me \$0.

Given the note about communication noise, it's possible:

- They actually sent more but noise reduced it to \$0 (seems unlikely for such a large reduction)
- Or they chose to send \$0, possibly because they were disappointed or misinterpreted something

However, the pattern shows:

- Round 6: They initiated restart with \$0.45
- Round 7: I responded with \$2
- Round 8: They sent \$0

This could be them defecting again, or it could be noise. Given the experimental design mentions noise, I should consider being somewhat forgiving. But I also need to protect myself from being exploited.

I think I should send a small amount to signal I'm still willing to cooperate but also show that I won't be taken advantage of. Sending \$1 seems reasonable - it's less than what I sent in Round 7 (showing some caution) but still shows willingness to cooperate.

usage:

```
{  
  "input_tokens": 622,  
  "output_tokens": 486,  
  "reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$24.720000000000002.

This round, The sender sent you \$0.85, so you received \$2.55. How much do you return to the sender? (0-2.55)

Assistant output:

Looking at the recent pattern:

Round 7: I sent \$2, they returned \$0
Round 8: They sent \$0
Round 9: I sent \$1, they returned \$0
Round 10: They sent \$0.85 (received \$2.55)

The sender has defected twice recently, returning nothing after receiving amounts. However, they are now sending again, which could signal a willingness to re-engage.

I'll return a modest amount that matches approximately what they sent, showing I'm willing to reciprocate proportionally but being cautious given the recent defections.

```
'''json
{"return": 0.90}
'''
```

Reasoning:

Let me analyze this situation:

Round 6: They sent \$0.45, I received \$1.35, I returned \$0.60
Round 7: I sent \$2, they received \$6, and based on my earnings going from \$20.72 to \$20.72, it seems they returned \$0
Round 8: They sent \$0, I received \$0, I returned \$0
Round 9: I sent \$1, they received \$3
Round 10: They sent \$0.85, I received \$2.55

My current total earnings are \$24.72. Let me track:

- After Round 8: \$20.72
- Round 9 as sender: \$5 - \$1 sent = \$4, plus whatever they return
- Round 10: I receive \$2.55

If my total is now \$24.72, that means:

- Starting Round 9: \$20.72
- Round 9: \$5 - \$1 = \$4, plus return
- After Round 9: Should be \$20.72 + \$4 + return
- Round 10: Add the base \$5 for this round

Wait, let me recalculate more carefully:

- After Round 8: \$20.72
- Round 9 I was sender: Started with \$5, sent \$1, so I have \$4 left, plus whatever they returned
- Round 10 I'm receiver: I start with \$5 base, receive \$2.55 from sender

Actually, the total earnings include everything accumulated. Let me think about what happened:

If after Round 8 I had \$20.72

Round 9: As sender, I get \$5, send \$1 (keeping \$4), and receive some return amount R9
Round 10: As receiver, I get \$5, receive \$2.55

So my total should be: $\$20.72 + \$4 + R9 + \$5 + \$2.55 = \$20.72 + \$11.55 + R9$

If my total is now \$24.72, then: $\$24.72 = \$20.72 + \$11.55 + R9$, which gives $R9 = -\$7.55$. That can't be right. I need to reconsider what the stated total represents-it's my balance before this current decision, so Round 9 must have already been completed and settled.

After Round 8 I had \$20.72. In Round 9 as the sender, I contributed \$1, keeping \$4, and received R9 in return. If my current balance is \$24.72, then $\$24.72 = \$20.72 + \$4 + R9$, meaning $R9 = \$0$. They sent nothing back. Looking at the pattern: Round 6 they sent \$0.45 and I returned \$0.60, Round 7 I sent \$2 and got nothing back, Round 8 they sent nothing, and now Round 9 they returned \$0 again. In Round 10 I'll receive the \$5 base plus \$2.55 from the sender's $\$0.85 \times 3$ multiplier, and I need to decide my return amount.

This sender has now defected in consecutive rounds (7 and 9), returning nothing despite receiving tripled amounts. The \$0.85 they're sending now is actually less than the \$1 I sent in Round 9. I'm inclined to return roughly what they sent-around \$0.85-to signal I'll reciprocate but won't tolerate exploitation....

```
Usage: {
  "input_tokens": 665,
  "output_tokens": 1273,
  "reasoning_tokens": 0
}
```